

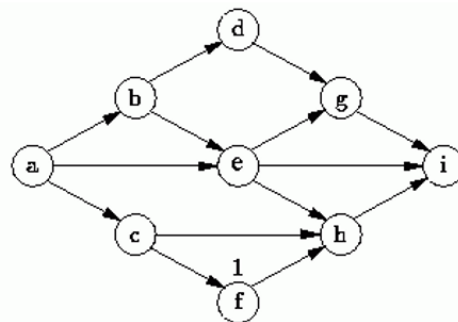
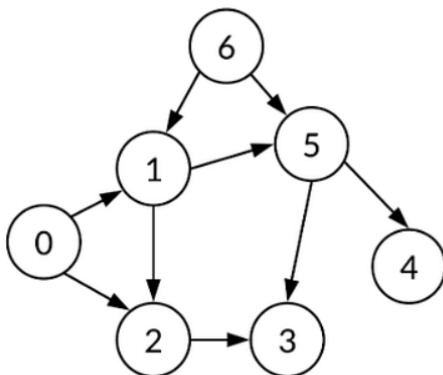
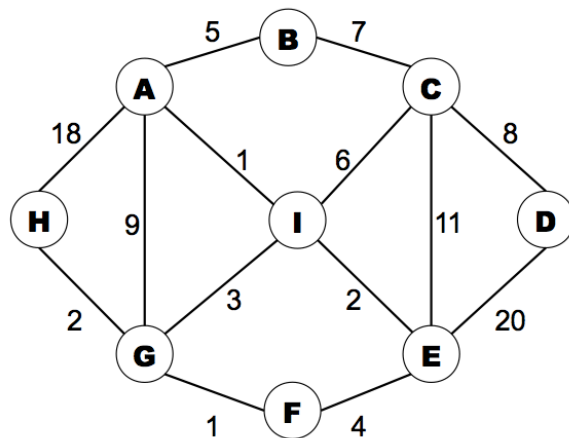
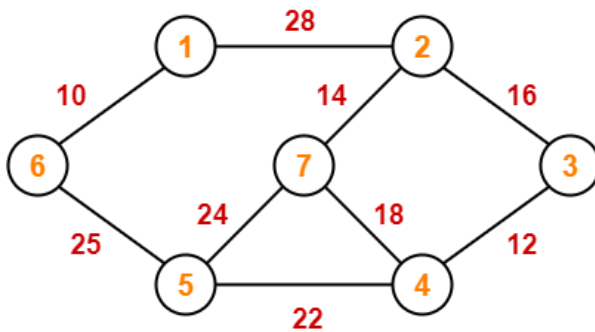
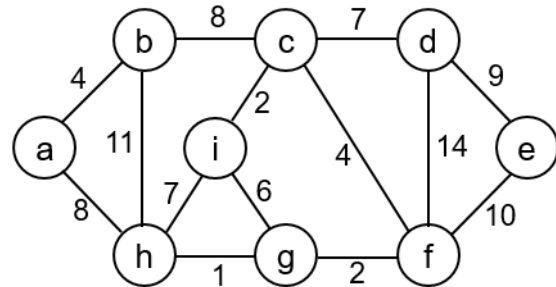
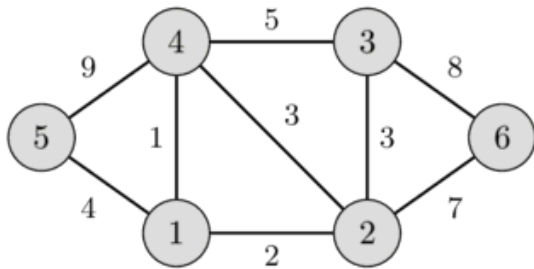
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IS43 Design and Analysis of Algorithms
Sec - B

1. Find the Optimal Chain Multiplication sequence for A (10*8) B(8*7) C(7*2)
2. Derive the LCS for the following strings CGAGCT ATGCAT
3. What is the Optimal BST for the following nodes and probabilities
A (0.15) B(0.30) C (0.10) D (0.20)
4. Calculate the Huffman Codes for the phrase "Huffman's Buffet"

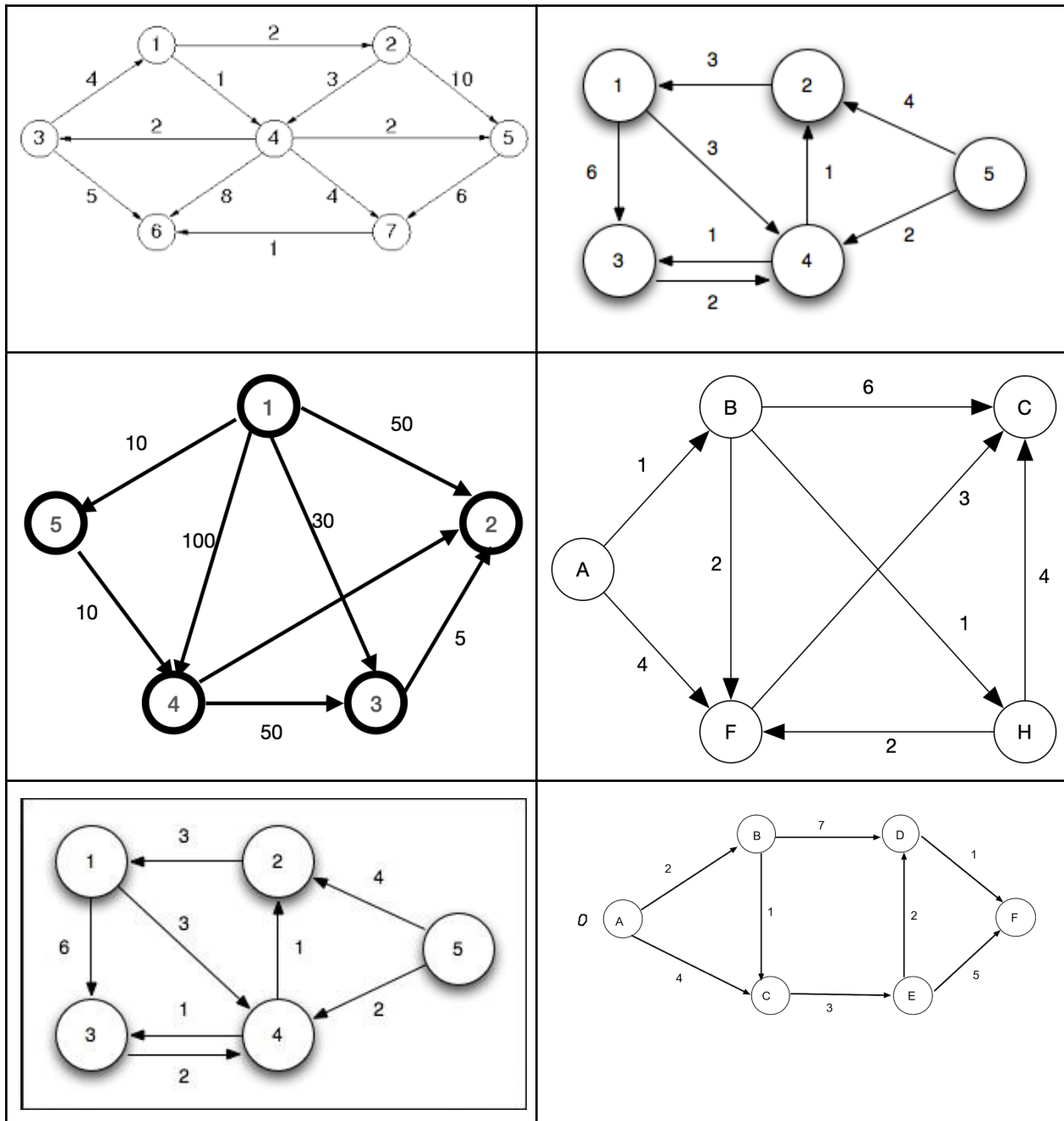
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IS43 Design and Analysis of Algorithms
Sec - B
MST & Topological Sorting



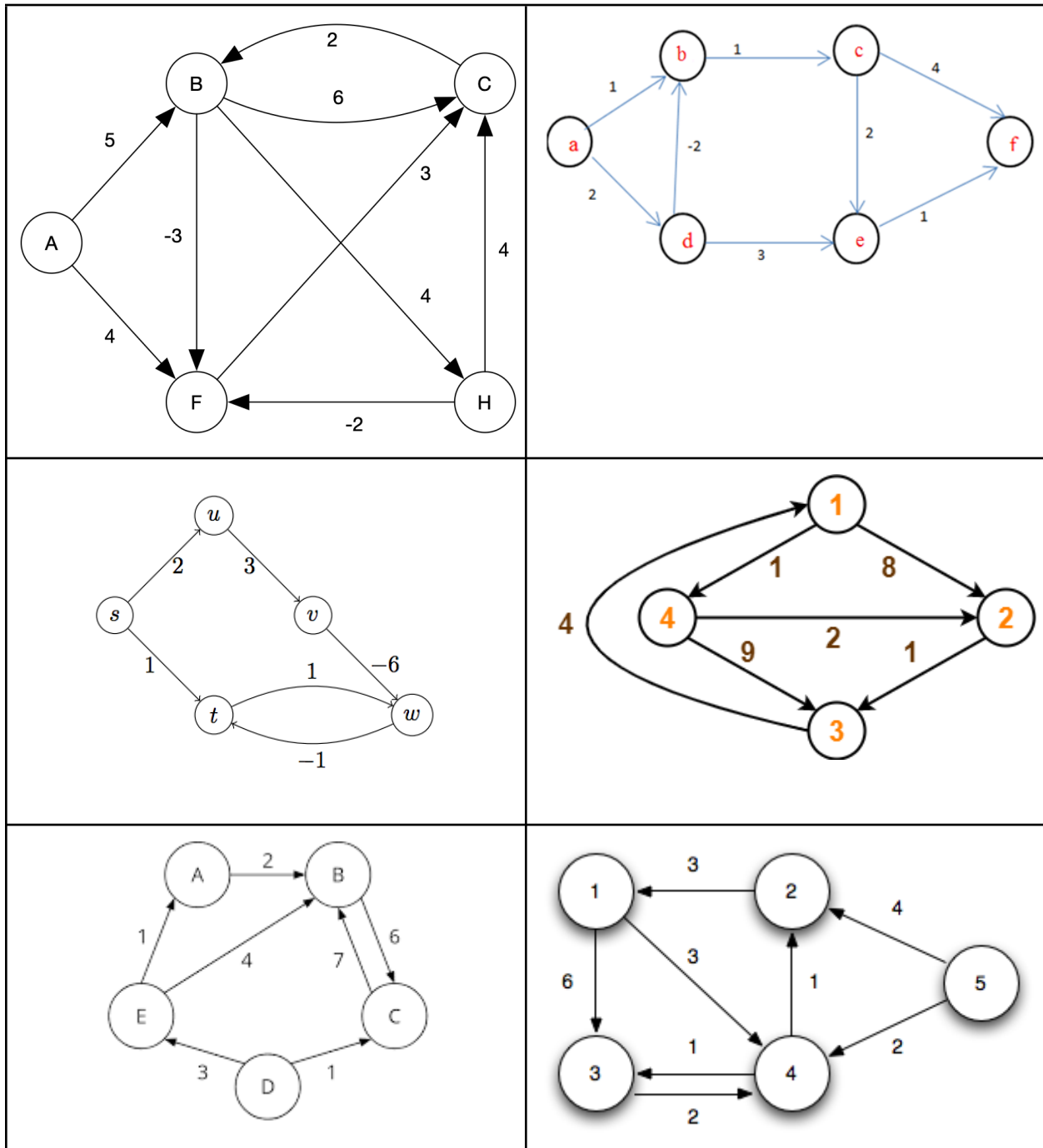
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IS43 Design and Analysis of Algorithms
Sec - B
Dijkstra shortest path



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IS43 Design and Analysis of Algorithms
Sec - B
Bellman Ford & Floyd Warshall



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IS43 Design and Analysis of Algorithms

Sec - B

1. Find the MST for the following graph (A, B, 10), (A, C, 12), (B, C, 9), (B, D, 8), (C, E, 3), (C, F, 1), (D, E, 7), (D, F, -1), (D, G, 8), (D, H, 5), (E, F, 3), (F, H, 6), (G, H, 9), (G, I, 2), (H, I, 11)
2. Apply Prim's Algorithm to find MST for the following graph (0, 1, 4), (0, 7, 8), (1, 2, 8), (1, 7, 11), (2, 3, 7), (2, 5, 4), (2, 8, 2), (3, 4, 9), (3, 5, 14), (4, 5, 10), (5, 6, 2), (6, 7, 1), (6, 8, 6), (7, 8, 7)
3. Apply Kruskal Algorithm to find MST for the following graph

	0	1	2	3	4	5	6
0	[0, 7, ∞ , 5, ∞ , ∞ , ∞],						
1	[7, 0, 8, 9, 7, ∞ , ∞],						
2	[∞ , 8, 0, ∞ , 5, ∞ , ∞],						
3	[5, 9, ∞ , 0, 15, 6, ∞],						
4	[∞ , 7, 5, 15, 0, 8, 9],						
5	[∞ , ∞ , ∞ , 6, 8, 0, 11],						
6	[∞ , ∞ , ∞ , ∞ , 9, 11, 0]]						

4. Find the Topological sort order for the graph (7, 1, 1), (7, 0, 1), (5, 1, 1), (3, 1, 1), (3, 4, 1), (3, 0, 1), (1, 4, 1), (1, 2, 1), (1, 6, 1), (0, 6, 1)
5. Find the shortest path from A for the given graph (A, B, 7), (A, C, 12), (B, C, 2), (B, D, 9), (C, E, 10), (E, D, 4), (E, F, 5), (D, F, 1)
6. Find the shortest path from 1 for the given graph (0, 1, 4), (0, 7, 8), (1, 2, 8), (1, 7, 11), (2, 3, 7), (2, 5, 4), (2, 8, 2), (3, 4, 9), (3, 5, 14), (4, 5, 10), (5, 6, 2), (6, 7, 1), (6, 8, 6), (7, 8, 7)
7. Find the shortest path from A for the given graph (A, B, 4), (A, F, 6), (B, C, -1), (B, E, 2), (B, F, -2), (C, D, 1), (C, E, 3), (E, D, -1), (F, E, -1)
8. Find all shortest paths for the given graph (A, B, 5), (A, D, 6), (B, C, 1), (B, E, 7), (C, A, 3), (C, D, 4), (D, C, 2), (D, E, 3), (E, A, 2), (E, D, 5)